

Challenge: Object Retrieval

Points: 350 • Estimated Time: 90 Minutes

Challenge Scope

The robot must pick up a designated tennis ball using a claw attachment and transport it across the room.

Instructions & Engineering Hints

- **Objective:** Design, build, and program a robot equipped with a claw attachment to locate, securely grip, lift, and transport a standard tennis ball from a designated starting zone to a target drop-off zone across the room.
- **Structural Considerations:** When designing your claw mechanism, consider the mechanical advantage needed to hold a sphere securely without crushing it or letting it slip during transit. Utilizing rubber bands or textured friction pads on the gripper fingers can greatly improve your hold.
- **Sensor Placement Tips:** Think about how your robot will detect the tennis ball. Placing an ultrasonic or color sensor strategically on the front can help the robot align itself accurately before attempting to grasp the object.
- **Strategic Programming:** Implement a closed-loop control or sequential state machine in your code. Break the challenge down into distinct phases: navigation to the object, reliable actuation of the claw, safe lifting (maintaining a low center of gravity to prevent tipping), and cross-room navigation to the destination.